PaddyGuard AI:

A Multimodal AI Based System for Rice Leaf Disease And Pest Detection, Treatment Recommendation And Knowledge Assistance



Project ID : R26-SE-015

Introduction

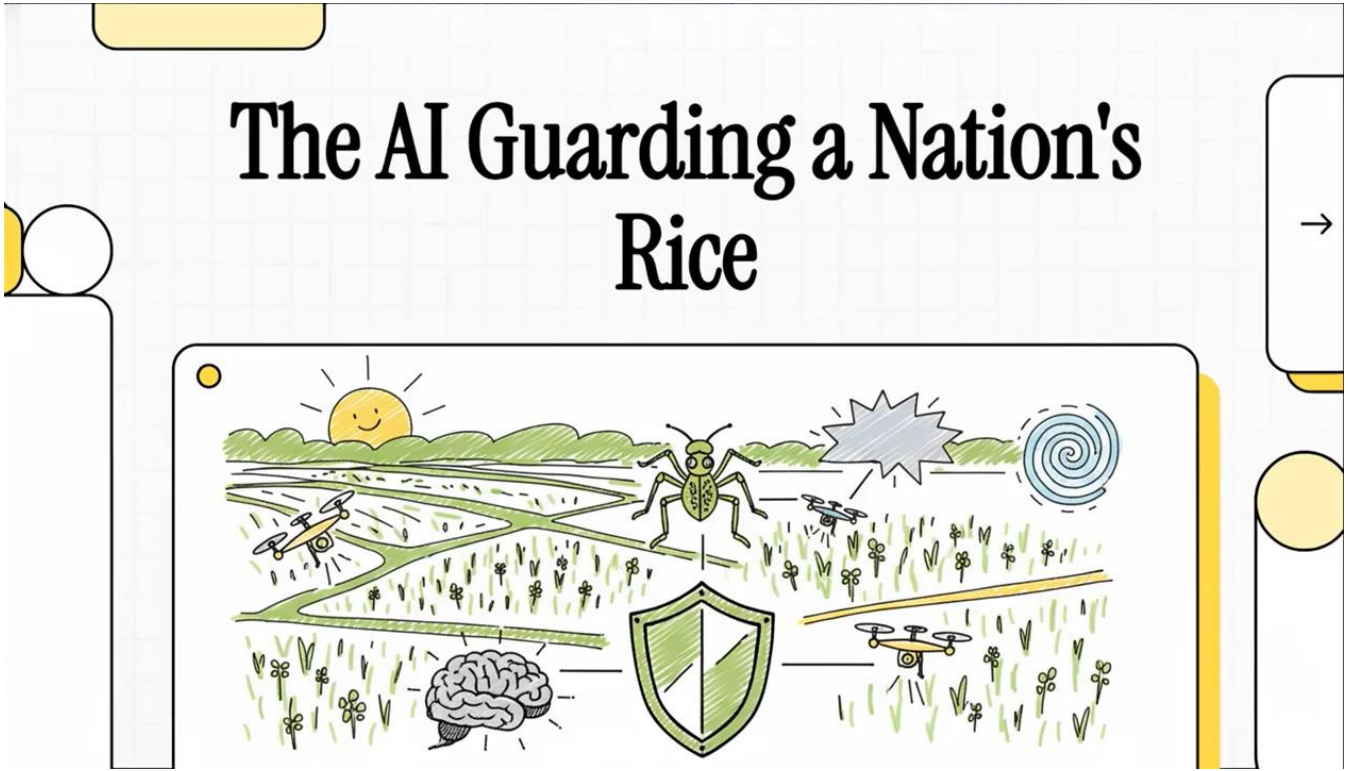
- Rice importance in Sri Lanka: feeds 21M people; 1.8M households rely on paddy farming.
- Challenges: diseases like Bacterial Blight, Rice Blast, Brown Spot can destroy 30% - 80% of harvest; limited access to experts.
- Existing tools limitations: mostly image-based, English-only, not farmer-friendly, require good cameras.

PaddyGuard AI solution:

- Voice & image-based disease detection.
- Treatment recommendations & pest identification.
- Sinhala-compatible advisory chatbot.
- Easy-to-use, mobile-friendly platform.
- Impact: supports SDG 2 (Zero Hunger) and SDG 9 (Innovation & Infrastructure) by protecting crops and empowering farmers.



Problem Definition





Problem

- Late Disease Identification – Farmers rely on experience or delayed expert advice, causing wrong treatments and 30% - 80% crop losses.
- English-Only Systems – Most tools support only English, making them difficult for Sinhala-speaking farmers.
- Image Dependency – Existing tools require clear leaf photos, but many farmers lack smartphones with good cameras.
- Fragmented Solutions – No single platform integrates disease detection, pest identification, and treatment advice.

Limitations of Existing Agricultural AI Systems

Plantix Mobile App



- Requires clear leaf images
- Limited local language support
- No voice interaction for farmers

PlantVillage / Nuru AI



- Image-based diagnosis only
- Requires internet connectivity
- Limited support for multiple farmer inputs

Agrio Crop Platform



- Commercial / Paid platform
- Designed mainly for large-scale farming
- Not affordable for small farmers

CNN-Based Rice Research



- Focused only on model accuracy
- No treatment advisory system
- Not a practical farmer application

Our Approach

Voice-Based Disease Diagnosis

- Farmers describe crop symptoms in Sinhala voice. The system converts speech to text, analyzes symptoms using NLP models, and provides voice and text diagnosis.

Image-Based Disease Classification

- Farmers upload leaf images. Deep learning models detect diseases such as Bacterial Blight, Rice Blast, and Brown Spot, with OOD detection to avoid incorrect predictions.

Treatment Advisory Chatbot

- A hybrid AI + knowledge-based system provides treatment recommendations, pesticide dosage guidance, and farmer-friendly advisory through a chatbot.

Rice Pest Detection

- YOLO/CNN models identify common rice pests from images, with OOD detection to flag unknown insects and provide proper treatment advice.





Solution Overview

What It Is?

- PaddyGuard AI is a mobile-friendly AI system for Sri Lankan paddy farmers.
- Combines voice diagnosis, image disease detection, pest detection, and treatment advisory.
- Designed with Sinhala voice-first support for easy use by rural farmers.

What It Does?

- Voice-based disease diagnosis using Sinhala speech.
- AI image detection for rice diseases and pests.
- Provides treatment advice and pesticide guidance.
- Results delivered in Sinhala text and voice.

Voice-Based Rice Disease Diagnosis Using Speech Recognition And Natural Language Processing



IT22273680 | Weerasinghe K G J P

Contribution of your component to the overall solution

Enables Sinhala-speaking paddy farmers to voice-diagnose rice diseases without cameras or technical expertise.

Novelty of your component

First Sinhala-language voice-based rice disease diagnosis using Whisper ASR + BERT/TF-IDF classification with adaptive follow-up questioning.

Problem And Research Objectives.

Problem Definition

- Existing AI crop disease systems rely on leaf photographs.
- Requires smartphones with good-quality cameras.
- Rural Sri Lankan farmers often lack such devices.
- Most agricultural AI systems support only English.
- Creates a significant barrier for Sinhala-speaking farmers.

Research Objectives.

Develop a Sinhala voice-based NLP disease classification system for low-literacy Sri Lankan paddy farmers achieving ≥85% accuracy.

- Create a Sinhala agricultural voice dataset.
- Design and implement a speech-to-text and NLP pipeline .
- Evaluate and compare different text classification models.
- Build integration API to sister components. (C2, C3, C4)

Result: Farmers struggle to get quick, accurate disease diagnosis — leading to delayed treatment and crop losses.

Methodology

Methodology

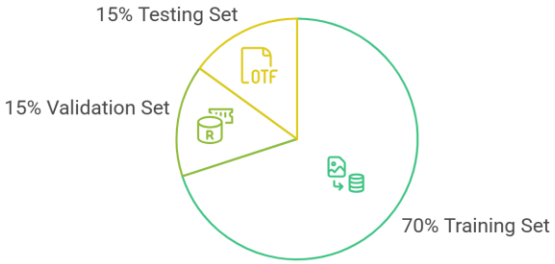
- **Speech Recognition** converts Sinhala voice input from farmers into text.
- **Natural Language Processing (NLP)** processes symptom descriptions and extracts important information.
- **Text classification models** analyze the symptoms to predict the most likely rice disease.
- The system learns relationships between **symptoms and diseases** using machine learning and transformer-based models.
- The final diagnosis is returned to the farmer using **text and voice output**.

Latest Technologies

- **Whisper Speech Recognition** – converts farmer voice into text.
- **TF-IDF + Machine Learning (SVM / Naive Bayes)** – baseline text classification.
- **BERT Transformer Model** – advanced contextual understanding of symptom descriptions.
- **PyTorch & HuggingFace Transformers** – model development and training.
- **Text-to-Speech (TTS)** – converts diagnosis results into Sinhala voice output.

Validation And Data Availability

Validation and Evaluation

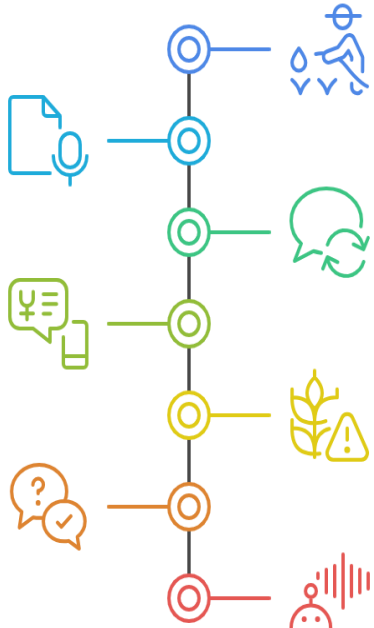


- Performance evaluated using accuracy, precision, recall, and F1-score.
- Compare performance between traditional ML models and transformer-based models.
- Conduct real voice input testing with farmers to evaluate system usability and prediction reliability.

Data Availability, Security Clearance

- Sinhala agricultural voice dataset will be **collected from farmers and volunteers**.
- Data will be stored securely with **restricted research access**.
- All API communications secured via HTTPS / TLS 1.3
- Conducted under SLIIT Research Ethics Committee guidelines.
- Participant privacy protected throughout data collection.
- Containerized microservices — Docker + Kubernetes horizontal scaling.

System Overview And Dependencies



Audio Capture and Validation

Microphone captures 16kHz WAV audio.

NLP Processing

Text processed for disease symptoms.

Dialogue Management

Follow-up questions asked if confidence is low.

User Interface

Rice Farmer records voice describing rice symptoms.

Automatic Speech Recognition (ASR)

Sinhala speech converted to text.

Disease Classification

Rice disease predicted from symptoms.

Output Generation

Voice response sent back to farmer.

Dependencies

- The predicted disease output from voice component used by the treatment recommendation component to provide advice to farmers.
- Common web User Interface (Web Application)

Common Components

- Shared agricultural datasets used for model training.
- Shared backend APIs for communication between components.
- Common data storage for symptoms, predictions, and farmer interactions.

Rice Leaf Disease Classification with Out-of-Distribution Detection

Hewanayake H.M.L.M | IT22168740



Contribution to Overall Solution

Classifies rice leaf diseases from images, filters out invalid inputs, and explains predictions visually.

Novelty

- Integrated OOD detection (ODIN, Mahalanobis, OpenMax) — rejects non-rice/unknown images.
- Grad-CAM visual explanations highlighting disease lesion regions.
- Multi-architecture CNN comparison: ResNet50, EfficientNetB3, DenseNet121.

Problem Definition & Research Objective

Problem Definition

- AI tools sometimes give wrong predictions for non-disease images.
- Wrong diagnosis → wrong pesticide use.
- Wastes money and harms the environment.
- Current tools lack image validation and clear explanations.
- Sri Lankan farmers need a reliable and understandable AI assistant.



Rice Leaf Disease — Field Sample

Research Objectives

- **Main Objective**: Build and test an AI model that correctly identifies rice leaf diseases (Bacterial Blight, Leaf Blast, Brown Spot, or Healthy) — and knows when to say "I don't recognize this image." Target: 92%+ accuracy, 85%+ rejection of invalid inputs.

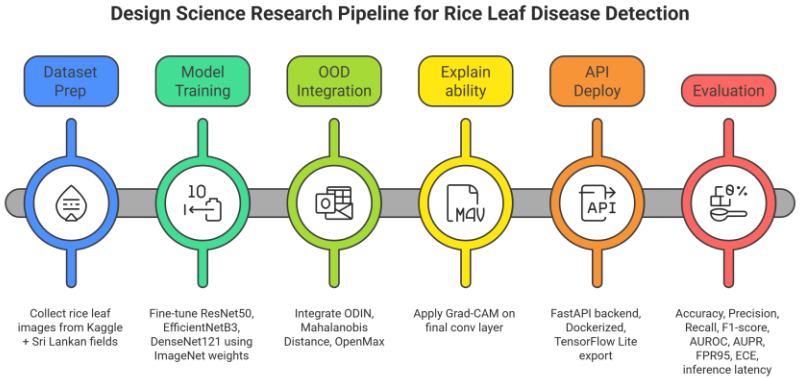
- **Sub-objectives**:
  - Compare three methods for detecting unknown/invalid images. (ODIN, Mahalanobis, OpenMax)
  - Show farmers *where* on the leaf the AI spotted the disease. (Grad-CAM heatmaps)

Methodology & Latest Technologies

Research Methodology:

DSR — Build → Test → Refine (6 Phases)

1. **Dataset Preparation** — Collect & clean rice leaf images
2. **Model Training** — Train 3 CNN models
3. **OOD Integration** — Add rejection capability
4. **Explainability** — Add Grad-CAM heatmaps
5. **API Deploy** — Package & deploy to mobile
6. **Evaluation** — Measure accuracy & performance



Latest Technologies:

- **Disease Detection (CNN)**
  - ResNet50, EfficientNetB3, DenseNet121
  - Transfer Learning from ImageNet
- **Unknown Image Rejection (OOD)**
  - ODIN, Mahalanobis Distance, OpenMax
  - No retraining needed
- **Explainability**
  - Grad-CAM — shows *where* disease is on the leaf
- **Deployment Stack**
  - FastAPI + Docker + TensorFlow Lite (mobile)
  - React.js frontend + MongoDB + TLS 1.3 security

Methodology & Latest Technologies

Validation & Evaluation

- Disease Classification — target above 92%
- Unknown Image Rejection— target above 85%
- Grad-CAM heatmaps validated by an agricultural domain expert to confirm disease region accuracy
- Inference latency benchmarked below 500ms for real-world mobile deployment

Data, Security & Ethical Clearance

Data Availability

- **Primary**: PlantVillage, Kaggle rice disease datasets. (public domain)
- **Secondary**: Locally collected Sri Lankan field images (farmer consent. obtained)
- OOD probe set: non-rice plants, insects, soil backgrounds, unseen diseases.

Security

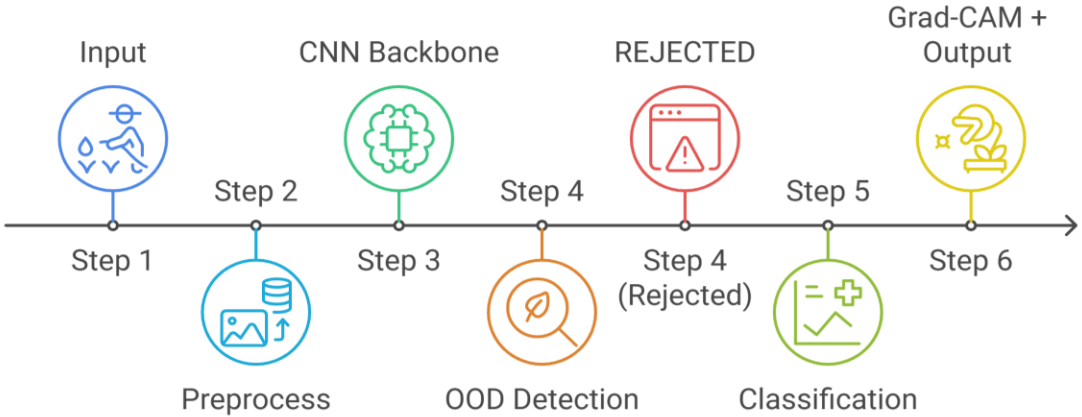
- TLS 1.3 encryption for all client-server communication.
- Images not permanently stored without explicit user consent.

Ethical Clearance

- No personally identifiable information (PII) collected in image dataset.
- Compliant with Sri Lanka Personal Data Protection Act. (PDPA)
- Approved by SLIIT Faculty Research Ethics Committee.

System Overview & Dependencies

System Overview of the Individual Component



Dependencies and Common Components

Component 1 — Voice Diagnosis
Component 1 and my component share the same backend server and database.

Component 3 — Treatment Chatbot
my component identifies the disease, it sends the result directly to Component 3, which then gives the farmer the treatment advice.“

Component 4 — Pest Detection
Component 4 and my component use the same mobile app screen and the same image upload button so the farmer only learns one interface.

A Multimodal AI System for Rice Pest Detection With Out-of-Distribution Detection



IT22065858 | KALHARA E P K

Contribution of component to the overall solution

Automatic detection of rice pests from images, helping farmers identify pest infestations early.

Novelty of component

Integration of deep learning–based pest detection with Out-of-Distribution (OOD) detection to identify.

Problem Definition & Research Objective

Problem Definition

- Rice pests destroy up to 40% of harvest if unmanaged.
- Small farmers don’t have access to agricultural experts.
- Misidentification leads to wrong pesticide use.
- Causes economic loss & environmental damage.
- Existing AI apps give wrong answers for unknown insects.

Research Objectives

Main Objective:

Develop a **deep learning–based system to detect rice pests from images accurately**

- Collect and prepare a rice pest image dataset for model training.
- Train and evaluate CNN and YOLO models for pest detection.
- Integrate Out-of-Distribution (OOD) detection to identify unknown insects.
- Evaluate model performance using accuracy, precision, recall.

Methodology & Latest Technologies

Methodology

- Use Deep Learning and Computer Vision techniques for pest image analysis.
- Train and compare CNN models for pest classification.
- Apply **image preprocessing and data augmentation** to improve robustness.
- Integrate Out-of-Distribution **(OOD)** detection to identify unknown insects.

Dataset Preparation -> Model Training -> OOD Intergration -> explainability -> API Deploy -> evaluation

Latest Technologies :

Disease Detection (CNN) Models
ResNet50, EfficientNetB0, MobileNetV2

Object Detection
YOLO v8 for small insects in field images

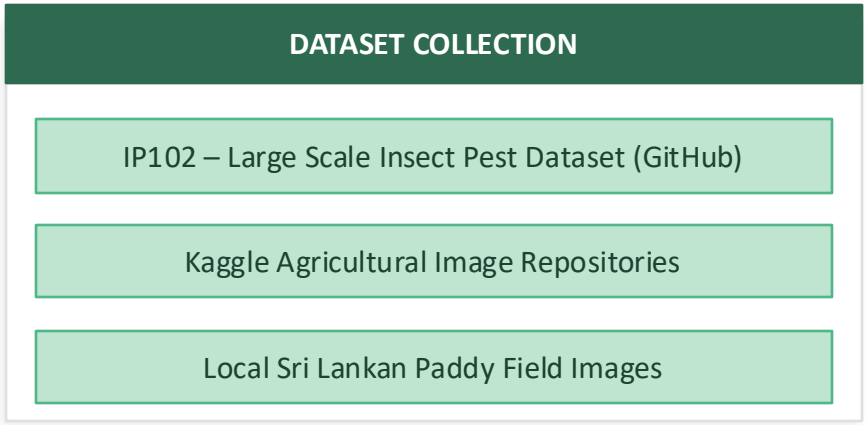
Data Augmentation
Rotation, flipping, brightness adjustment

Unknown image detection (OOD Detection)
ODIN / Mahalanobis / OpenMax

Tools
Python / PyTorch / Docker / Google Colab GPU/mongoDB

Data Collection & Security, Ethical Considerations

DATASET COLLECTION



Dataset Split:



Data Security

Image data processed securely within the PaddyGuard AI system using protected cloud environments and secure API communication (HTTPS).

Ethical Considerations

Only publicly available datasets are used with proper acknowledgment, and no personal or sensitive user data is collected.

System Overview & Dependencies

System Overview

1. Farmer uploads pest image JPEG/PNG
2. Image preprocessed (OpenCV)
3. Deep learning model analyzes
4. OOD check: known or unknown?
5. Result shown: pest name + score & Clasification

Dependencies

- Shared dataset preparation.
- Google Colab GPU for model training.
- Integration with PaddyGuard AI system.

common components

- Shared agricultural datasets used for model training.
- Shared backend APIs for communication between components.
- Common data storage for symptoms, predictions, and farmer interactions.

Treatment Advisory System and Agricultural Chatbot



IT22303820 | Keshan B K

Contribution of your component to the overall solution

Provides farmers with accurate treatment recommendations, pesticide dosage guidance, and preventive farming advice after rice disease or pest diagnosis using a knowledge-driven advisory system.

Novelty of your component

Introduces a RAG-based agricultural chatbot integrated with agricultural knowledge sources (IRRI, FAO) and rule-based validation to deliver safe, context-aware, and explainable treatment recommendations.

Problem And Research Objectives.

Problem Definition

- Existing agricultural AI systems mainly focus on disease detection rather than treatment guidance.
- Farmers must search multiple sources to find the correct treatment methods.
- Incorrect pesticide usage can cause crop damage and environmental risks.
- Many systems provide generalized recommendations without considering crop conditions.

Research Objectives.

Develop a knowledge-driven treatment advisory system and agricultural chatbot that provides accurate treatment recommendations for rice diseases and pest infestations.

- Build a structured agricultural knowledge dataset from IRRI, FAO, and PlantVillage sources.
- Design and implement a knowledge-based treatment recommendation module.
- Develop an NLP-based agricultural chatbot for farmer queries.
- Integrate RAG to generate accurate advisory responses.

Result: Farmers receive reliable treatment guidance and preventive farming advice, enabling faster and safer crop management decisions.

Methodology

Methodology

- Farmers submit queries or receive diagnosis results from disease or pest detection modules.
- **Natural Language Processing (NLP)** processes farmer queries and identifies important agricultural keywords.
- **Semantic search and knowledge retrieval** are used to find relevant treatment information from the agricultural knowledge base.
- A **rule-based advisory engine** validates treatment recommendations based on agricultural guidelines.
- The final **treatment advice and preventive farming guidance** are delivered to farmers through the chatbot interface.

Latest Technologies

- **React.js Web Interface** – allows farmers to interact with the advisory chatbot.
- **Node.js & Express.js Backend API** – manages system communication and query processing.
- **LangChain / HuggingFace Transformers** – used for natural language processing and chatbot development.
- **Retrieval-Augmented Generation (RAG)** – retrieves relevant agricultural knowledge to generate accurate responses.
- **MongoDB Atlas** – stores agricultural knowledge datasets and treatment guidelines.

Validation And Data Availability

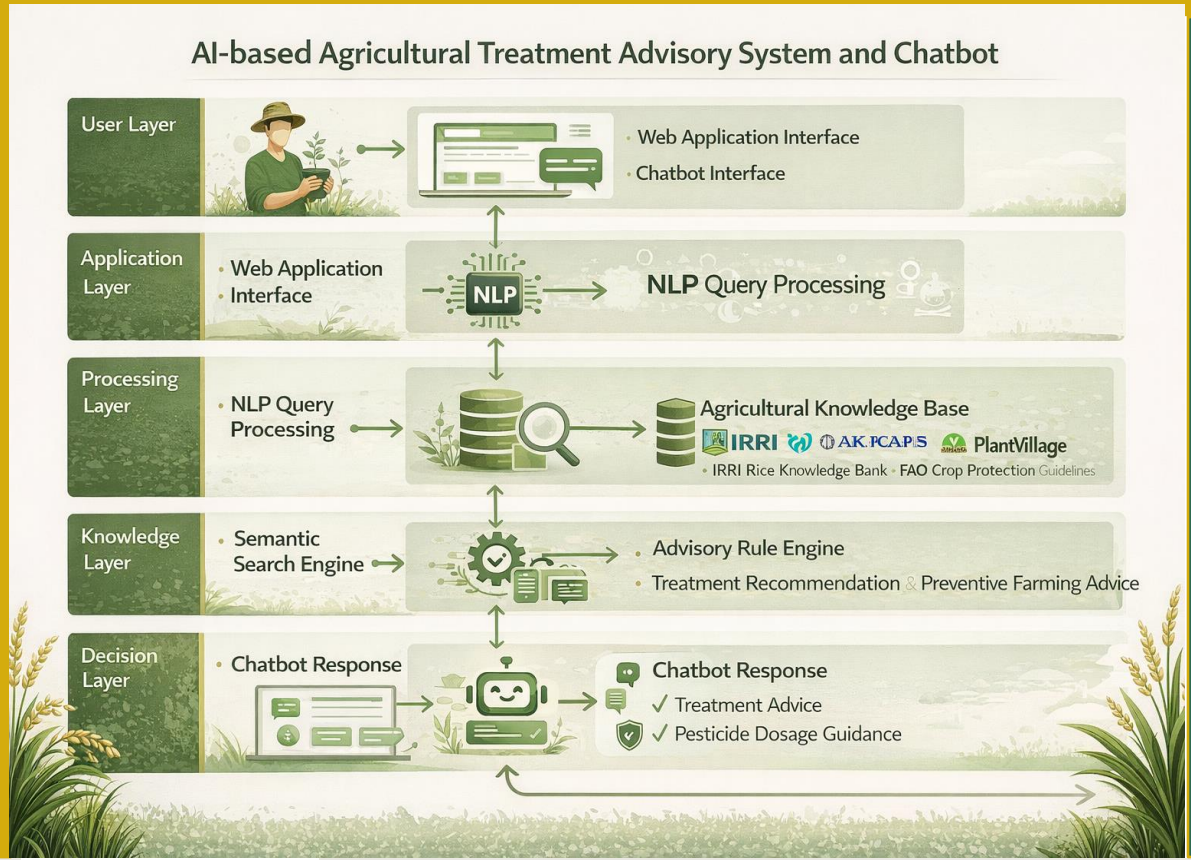
Validation and Evaluation

- System performance evaluated using response accuracy, recommendation relevance, and response timeCompare performance between traditional ML models and transformer-based models.
- Compare chatbot responses with agricultural guidelines from IRRI and FAO knowledge sources.
- Conduct query testing using predefined agricultural questions related to rice diseases and pest management.
- Evaluate the usability of the chatbot interface to ensure farmers can easily understand treatment recommendations.

Data Availability, Security Clearance

- Agricultural knowledge datasets collected from IRRI Rice Knowledge Bank, FAO Crop Protection Guidelines, and PlantVillage metadata.
- Knowledge base stored securely using a **MongoDB database with restricted research access**.
- All system communications secured through **HTTPS and secure API protocols.**
- Conducted under SLIIT Research Ethics Committee guidelines.
- Farmer queries used for evaluation will be **anonymized to protect user privacy.**

System Overview And Dependencies



Dependencies

- **Rice Disease Detection Module** provides disease prediction results to the advisory system.
- Rice Pest Detection Module sends pest identification outputs for treatment recommendation.
- **Voice-Based Symptom Analysis Module** supplies symptom descriptions from the farmer's voice input.
- **Backend API Server** connects the advisory system with other PaddyGuard AI components.

Common Components

- Shared agricultural datasets used for model training.
- Shared backend APIs for communication between components.
- Common web UI & data storage for symptoms, predictions, and farmer interactions.

Market, Value Proposition And Commercialization

Value Proposition

- Farmers: Instant Sinhala voice-based disease diagnosis & treatment advice.
- Government: Scalable digital advisory for rural farmers.
- Agri Companies: Targeted product recommendations at diagnosis stage.
- Society: Reduced crop losses and improved food security.

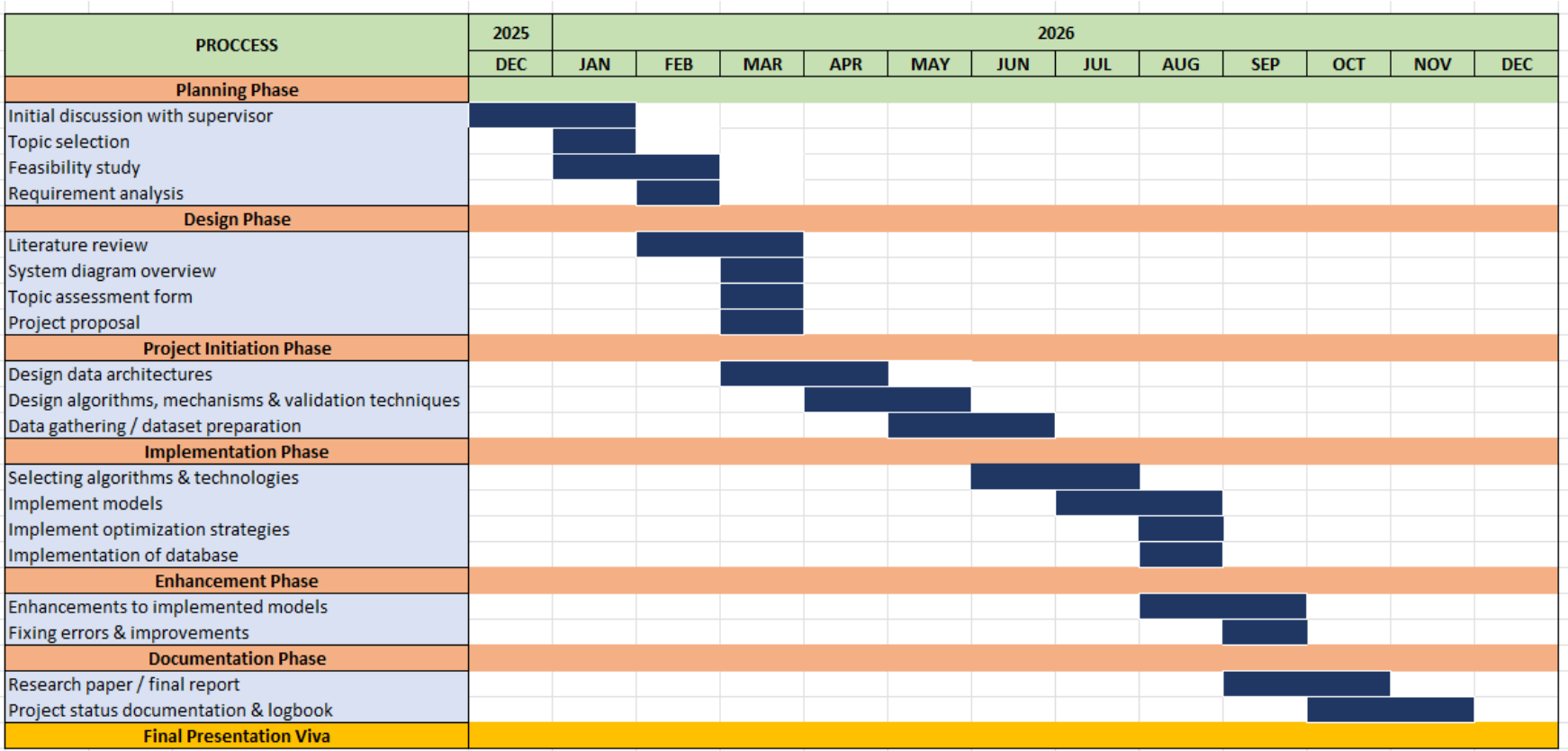
Target Market

- Primary: Department of Agriculture & extension services.
- Secondary: Agricultural cooperatives, crop insurance, agri-input companies.
- End Users: ~1.8M paddy farming households in Sri Lanka.

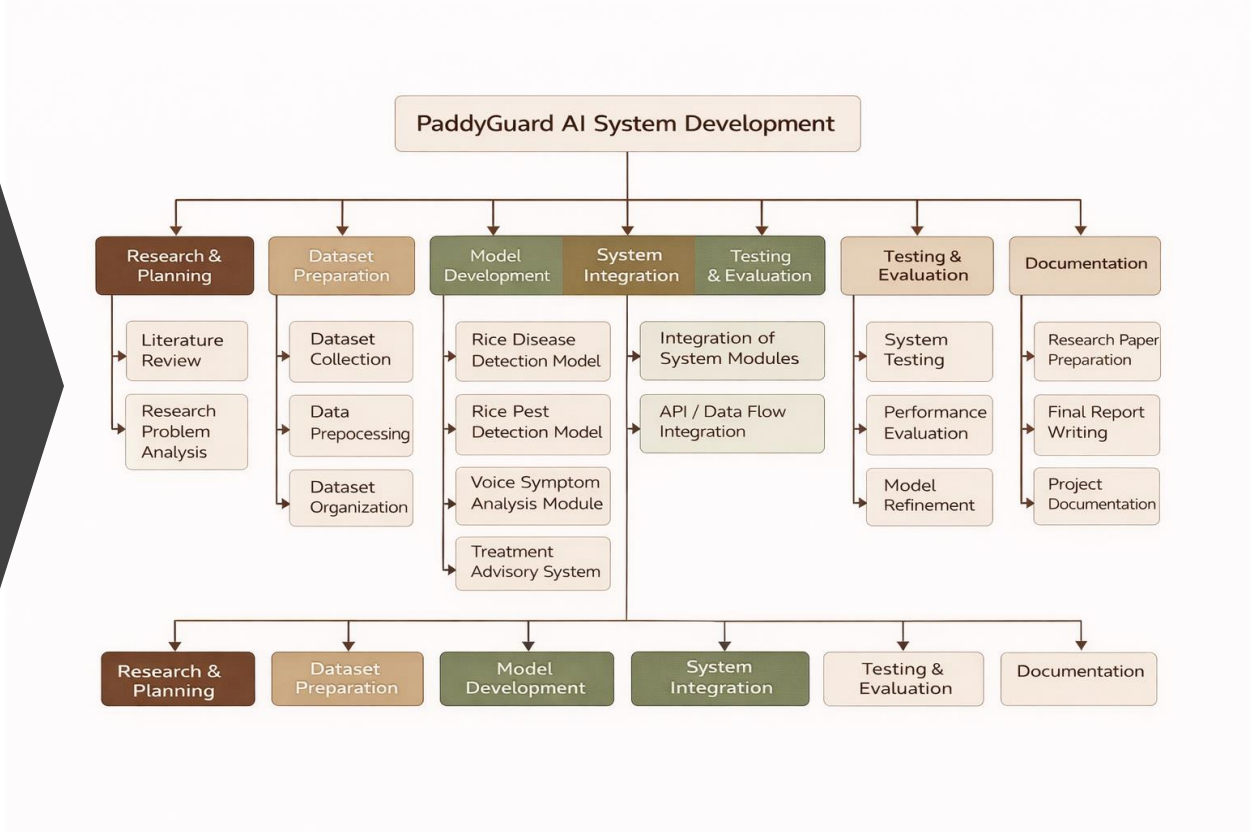
Commercialization

- Government licensing as national agricultural advisory tool.
- API integration for agri-input companies.
- Grants & SaaS analytics for cooperatives and regional expansion.

Gantt Chart


PROCCESS
2025
2026
DEC
JAN
FEB
MAR
APR
MAY
JUN
JUL
AUG
SEP
OCT
NOV
DEC
Planning Phase
Initial discussion with supervisor
Topic selection
Feasibility study
Requirement analysis
Design Phase
Literature review
System diagram overview
Topic assessment form
Project proposal
Project Initiation Phase
Design data architectures
Design algorithms, mechanisms & validation techniques
Data gathering / dataset preparation
Implementation Phase
Selecting algorithms & technologies
Implement models
Implement optimization strategies
Implementation of database
Enhancement Phase
Enhancements to implemented models
Fixing errors & improvements
Documentation Phase
Research paper / final report
Project status documentation & logbook
Final Presentation Viva

Work breakdown structure





Thank You